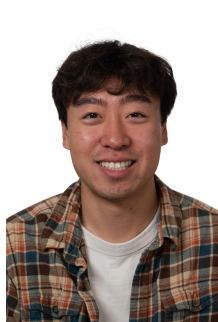




Chirp: Distributed Radar Network

Our Team



Jason Wang
MQTT Development



Oviya Seeniraj
Time Sync & Real-Time
Adaption



Andrey
Otvagin
Jetson/Node
Development



William Ni
Integration



Vihan Jayaraman
Wireless Infrastructure

Seeing without Light: RF Sensing

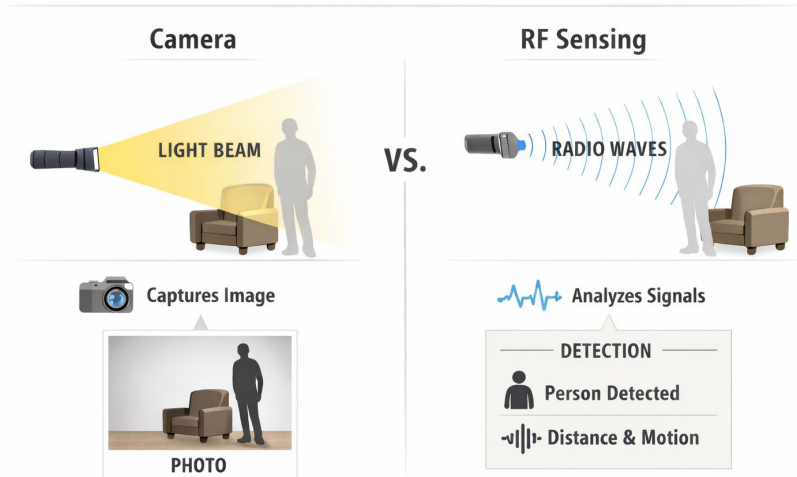


Fig. Camera vs RF Sensing

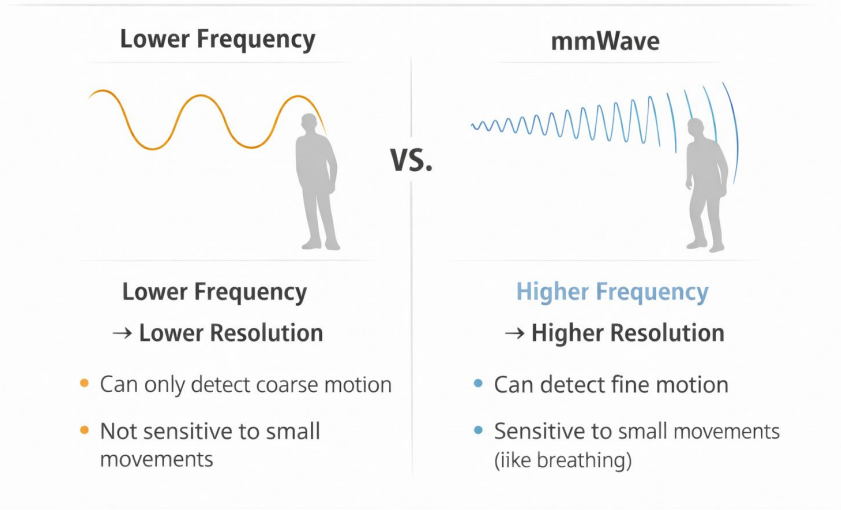
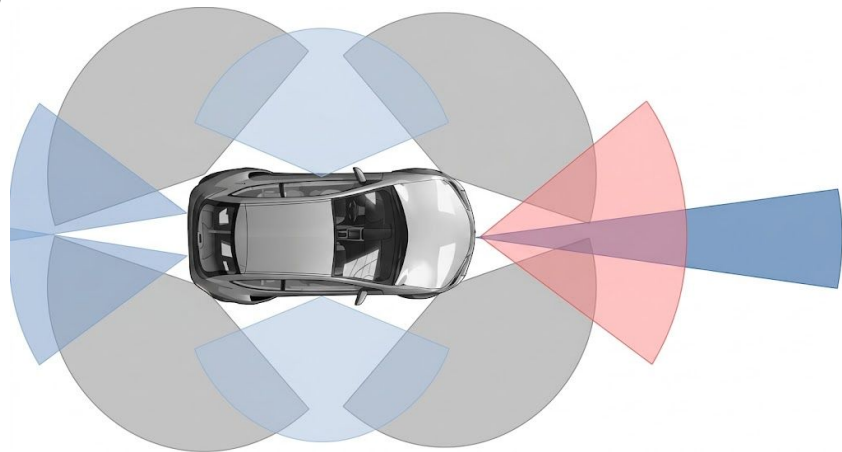


Fig. Sensing in mmWave vs Lower Frequency bands

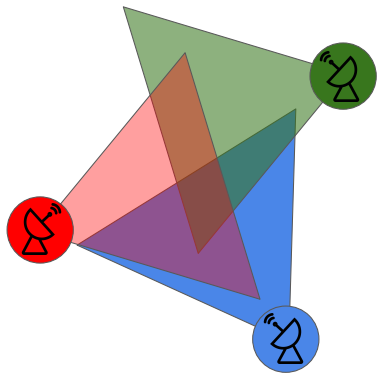
Motivation: The Problem

- Real time trajectory tracking is vital to safety in the age of automation
 - Car collisions
 - Non-intrusive fall detections in hospital
 - Robotics
- Radars have limited FOVs and ranges
- Real-time sensor fusion to generate larger maps

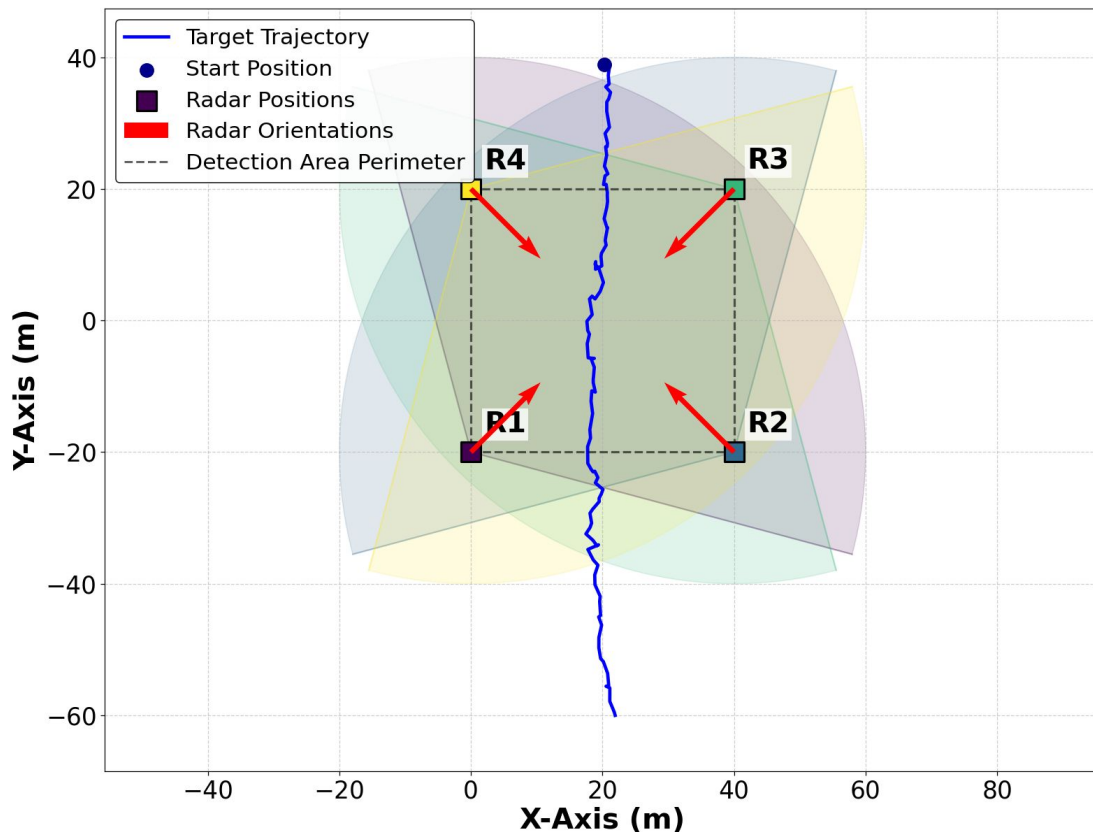


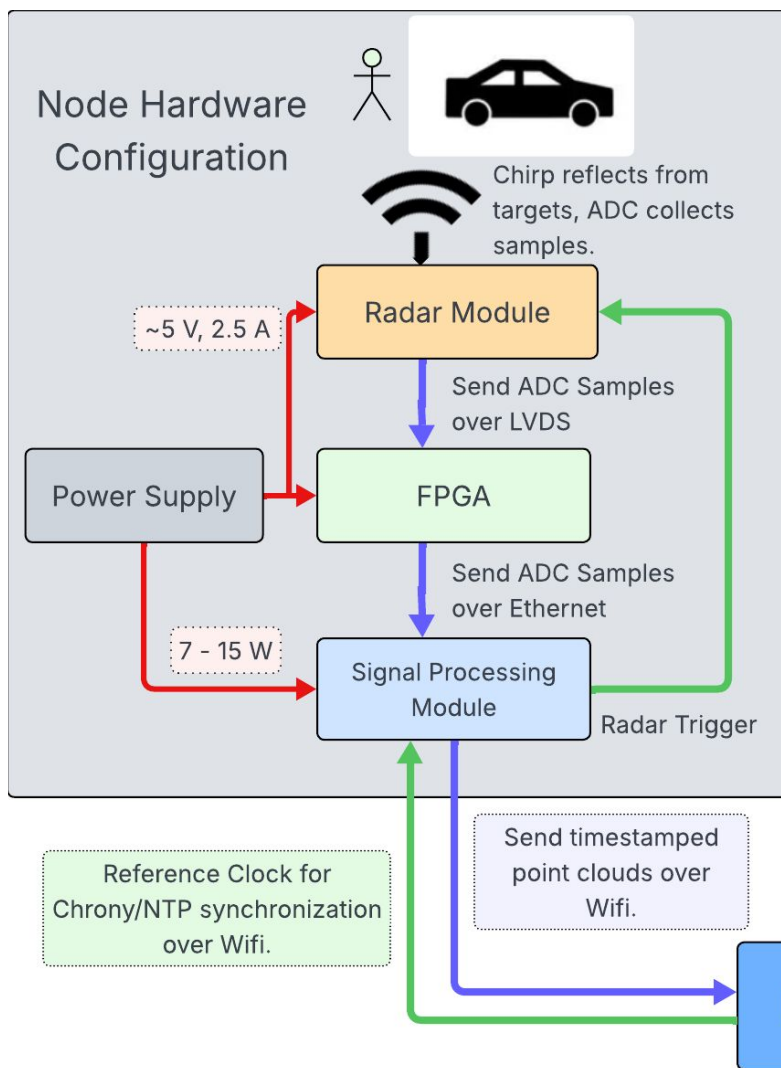
Goal & Purpose

- Demonstrate *Spatial Calibration* for distributed radar nodes
- Achieved through 3 main steps:
 - Time Synchronization
 - Reference Tracking
 - Relative Pose Estimation

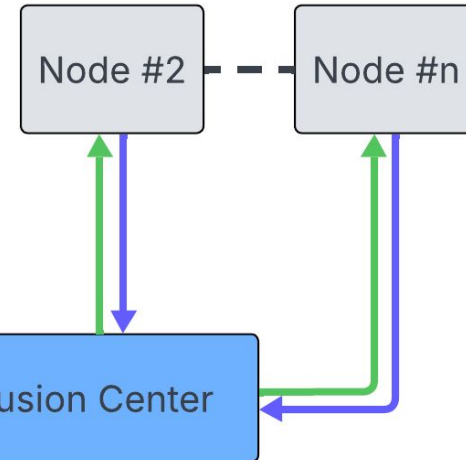
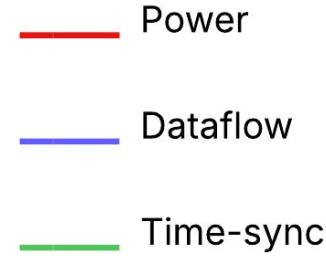


Ground Truth: 4-Radar SQUARE Configuration

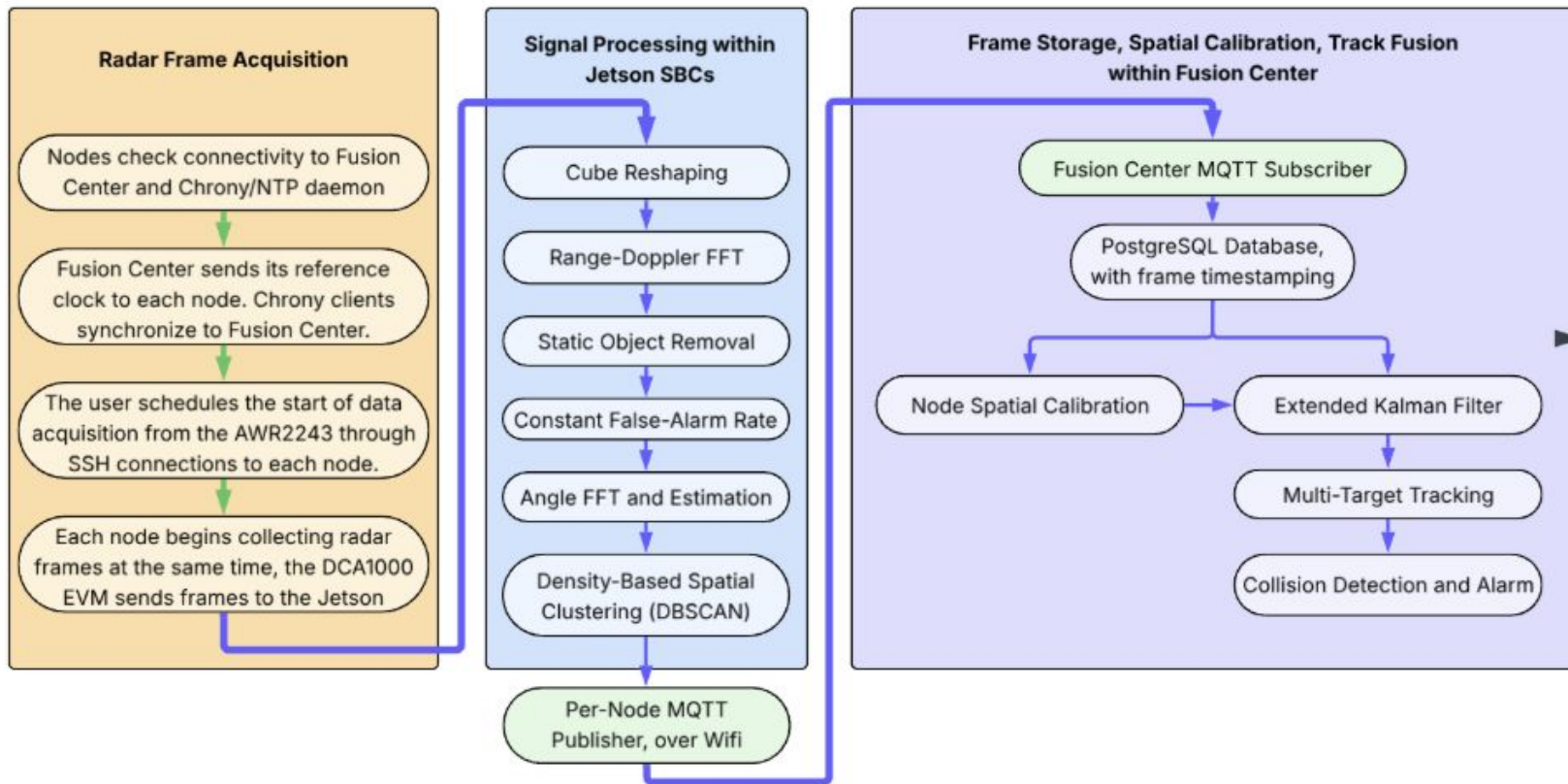




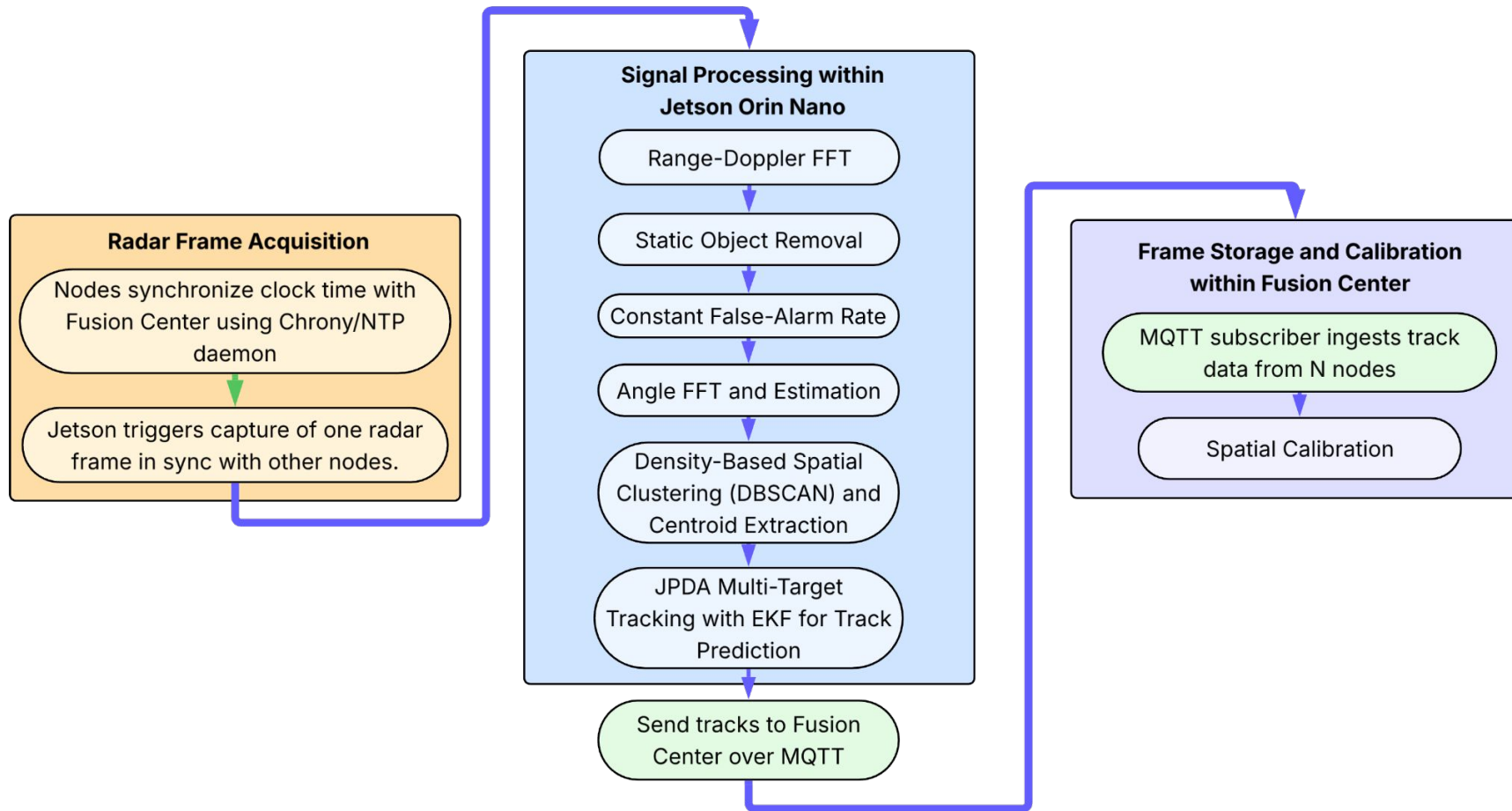
Legend



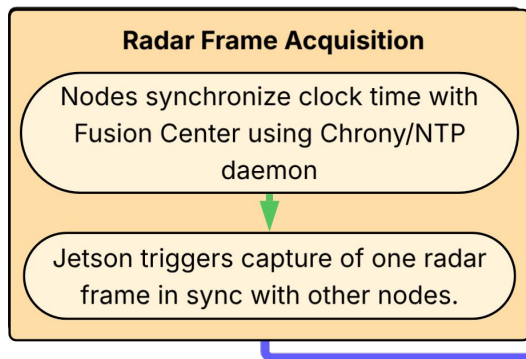
OLD DIAGRAM (ignore this slide)



Dataflow Overview

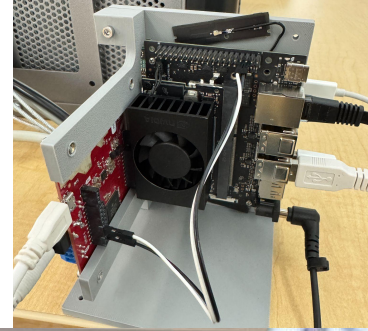


Dataflow Overview: Acquiring Radar Frames



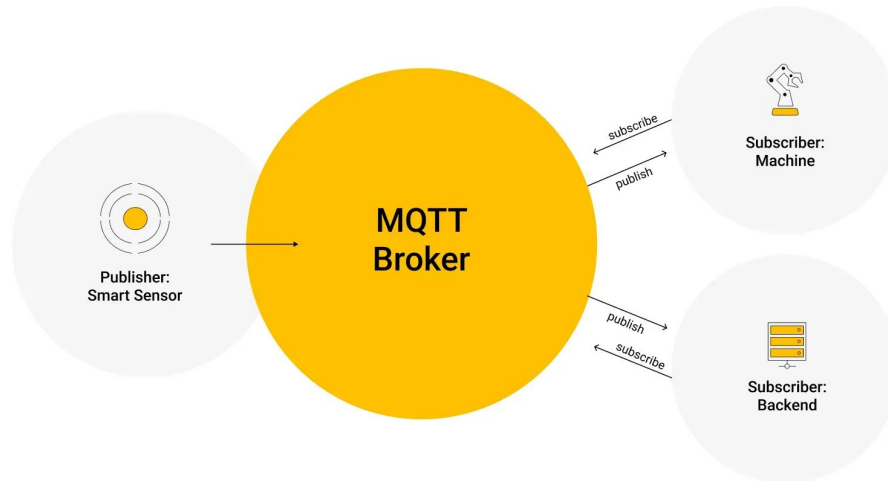
Time Synchronization through Chrony NTP Daemon

- Spatial Calibration requires that nodes capture frames at the same time
 - Synchronize Nodes
 - Setup Radars to capture a frame via a GPIO pulse
- Chrony NTP server served off of Fusion Center
- ~ 3 ms accuracy

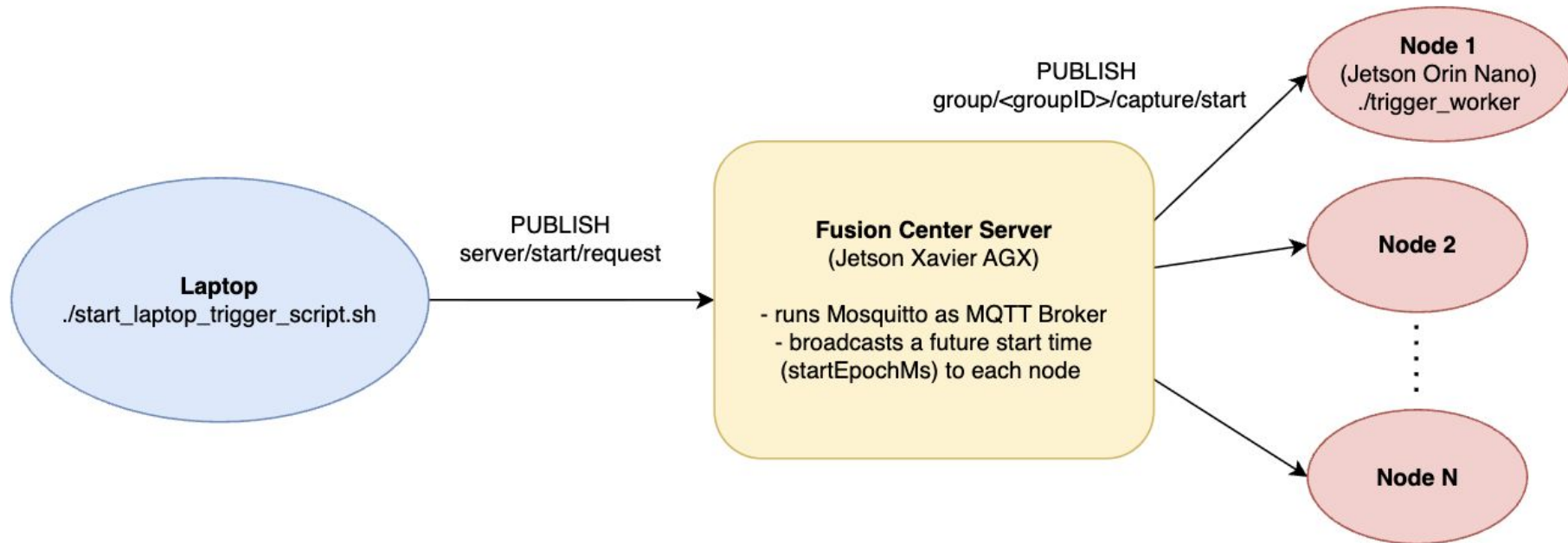


MQTT Overview for Communication between Nodes

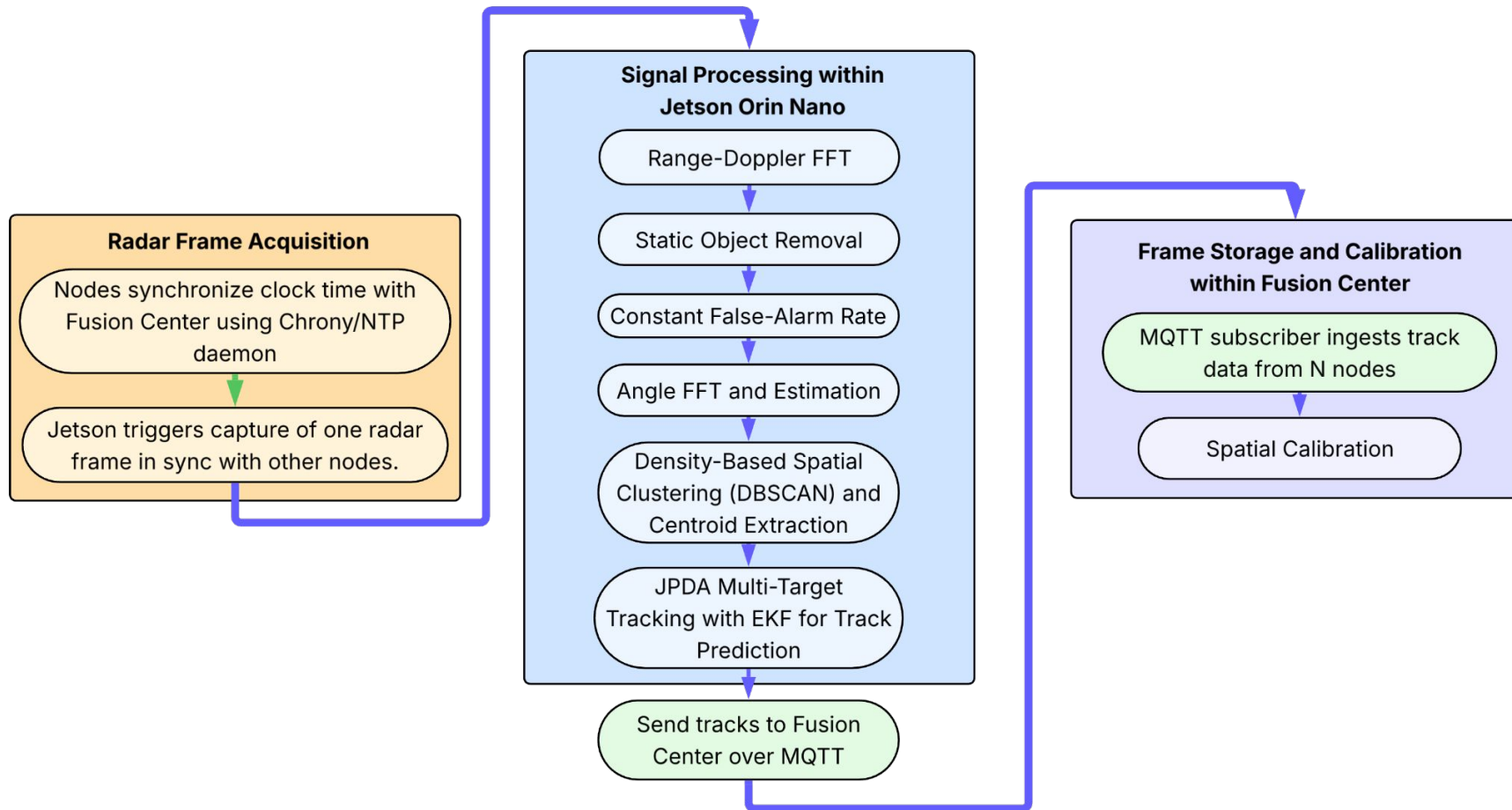
- MQTT = Message Queueing Telemetry Transport; common in IoT
- Publisher/subscriber model, where each node subscribes to “topics”
 - MQTT Broker handles routing of messages from publishers to subscribers



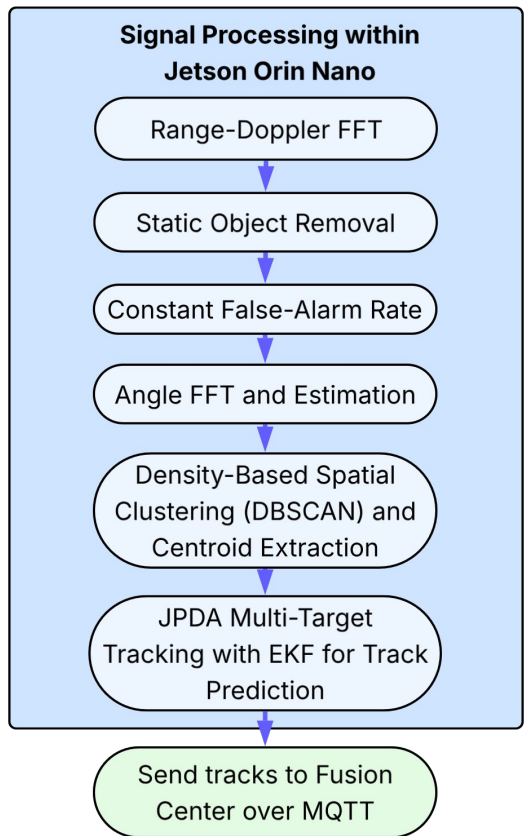
MQTT for Synchronized Trigger from Laptop



Dataflow Overview: Processing Radar Data

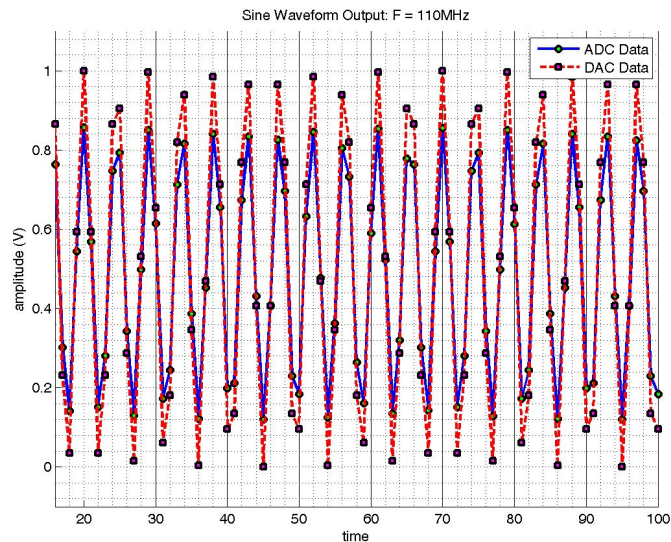


Dataflow Overview: Processing Radar Data



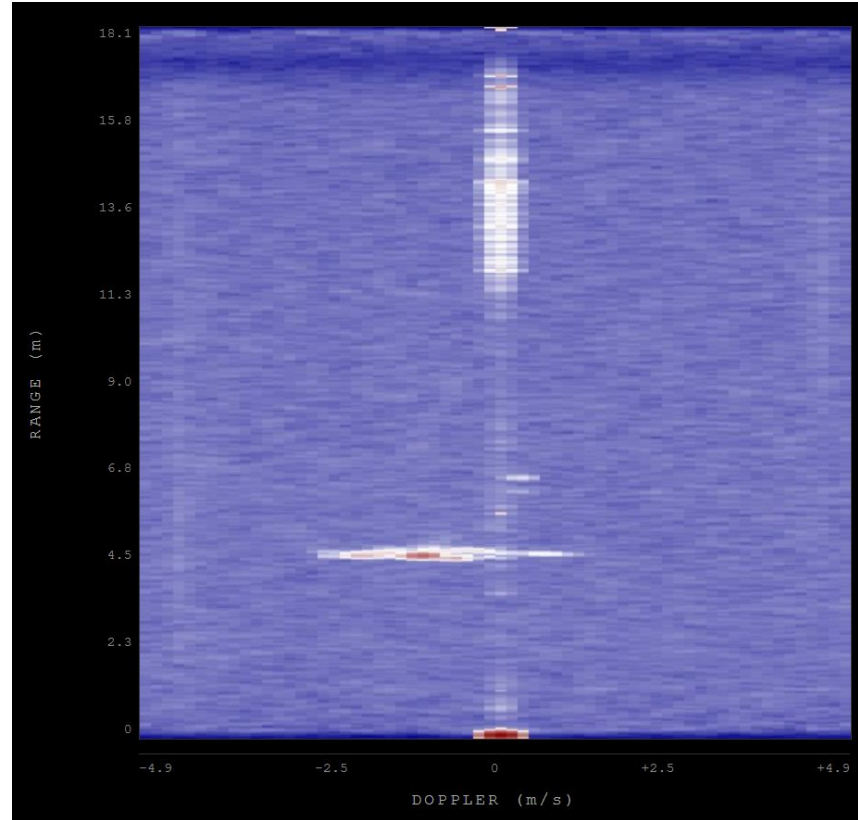
Radar Signal Processing Pipeline Purpose

- Stream of Raw ADC samples □ objects
location and velocity with respect
- Ex: Human 1
 - x: 1 m
 - y: 3 m
 - v_x: 1 m/s
 - v_y: 0.5m/s
 - t: 12h:34m:56s



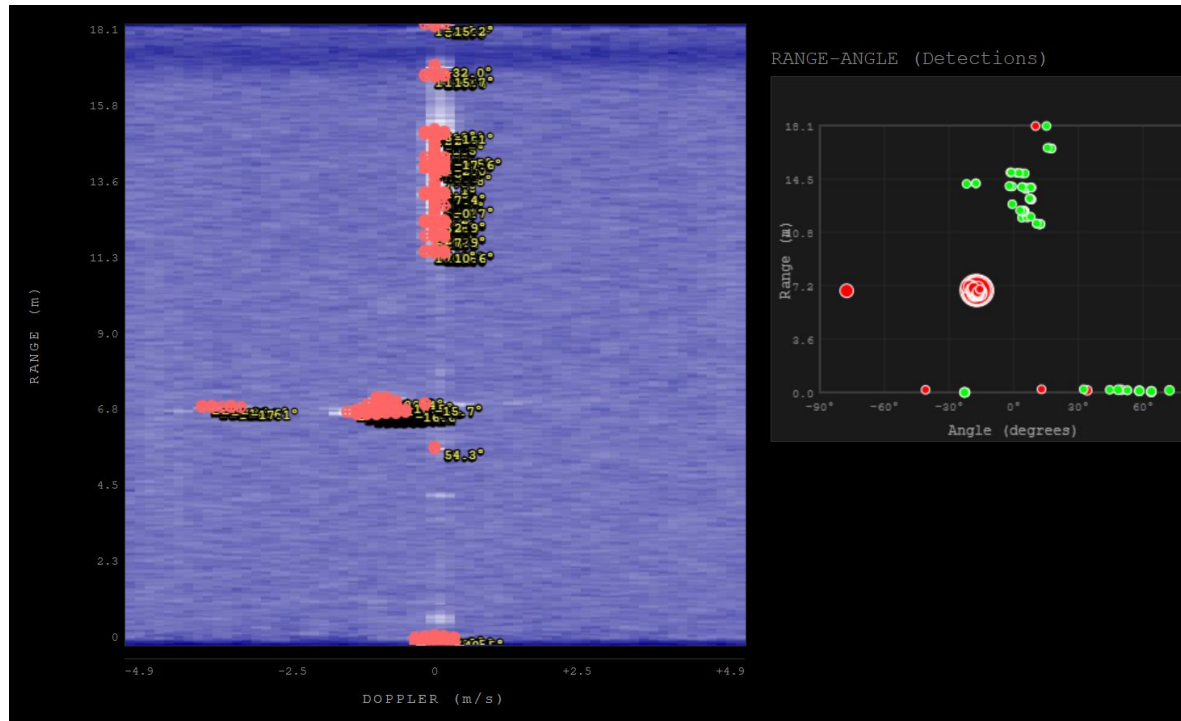
Pipeline: Raw Data to Range Doppler Map

- Data is provided as a stream of raw ADC samples
- 2D FFT to get a Range Doppler Map
- Subtract power of stationary background to only get moving targets



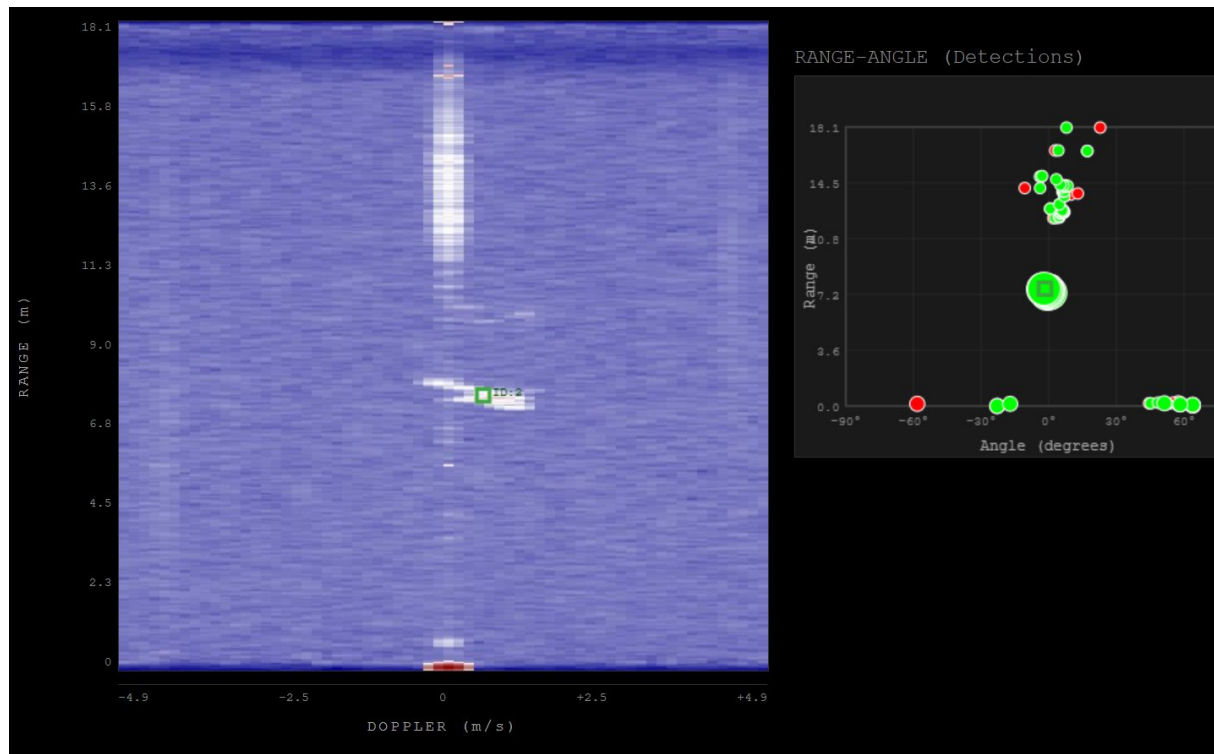
Pipeline: CFAR for Points of Interest

- CFAR to identify potential points of interest
- Calculate the angles of each of those points of interest



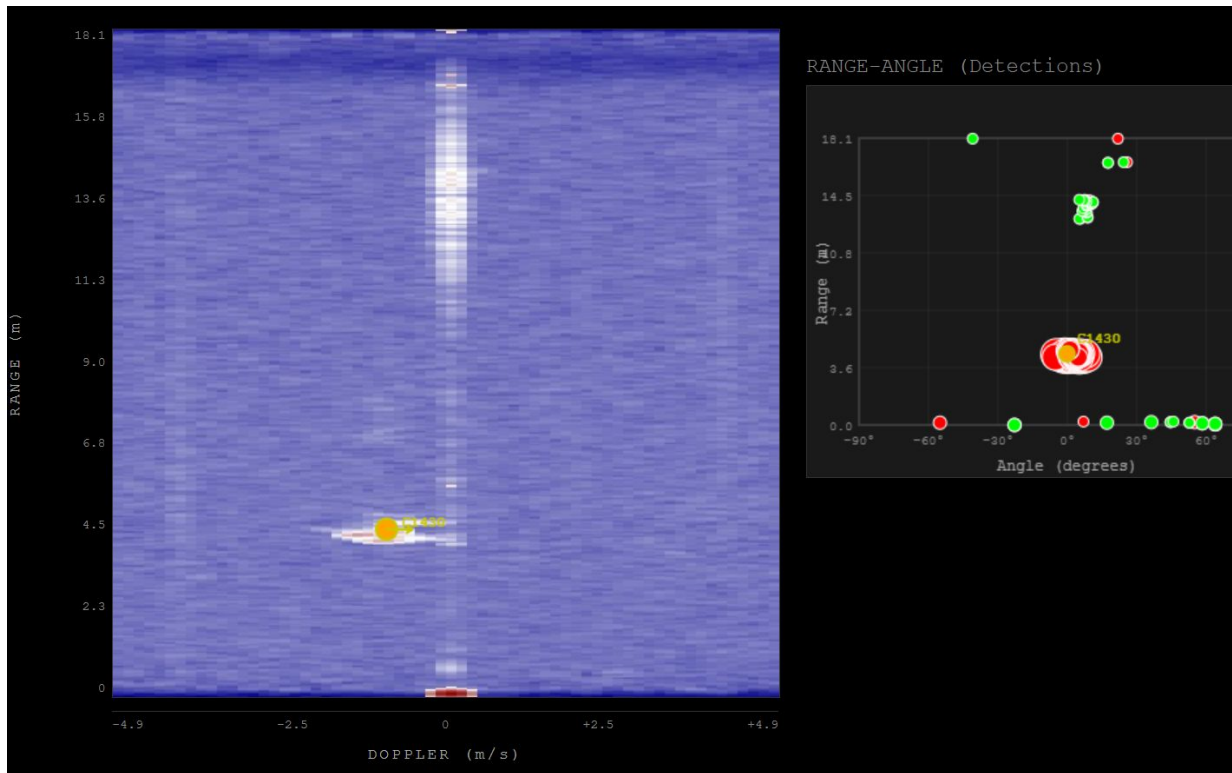
Pipeline: CFAR Point Clustering into Centroid

- Cluster CFAR Points (Range, Doppler, Angle) into Centroids using DBScan

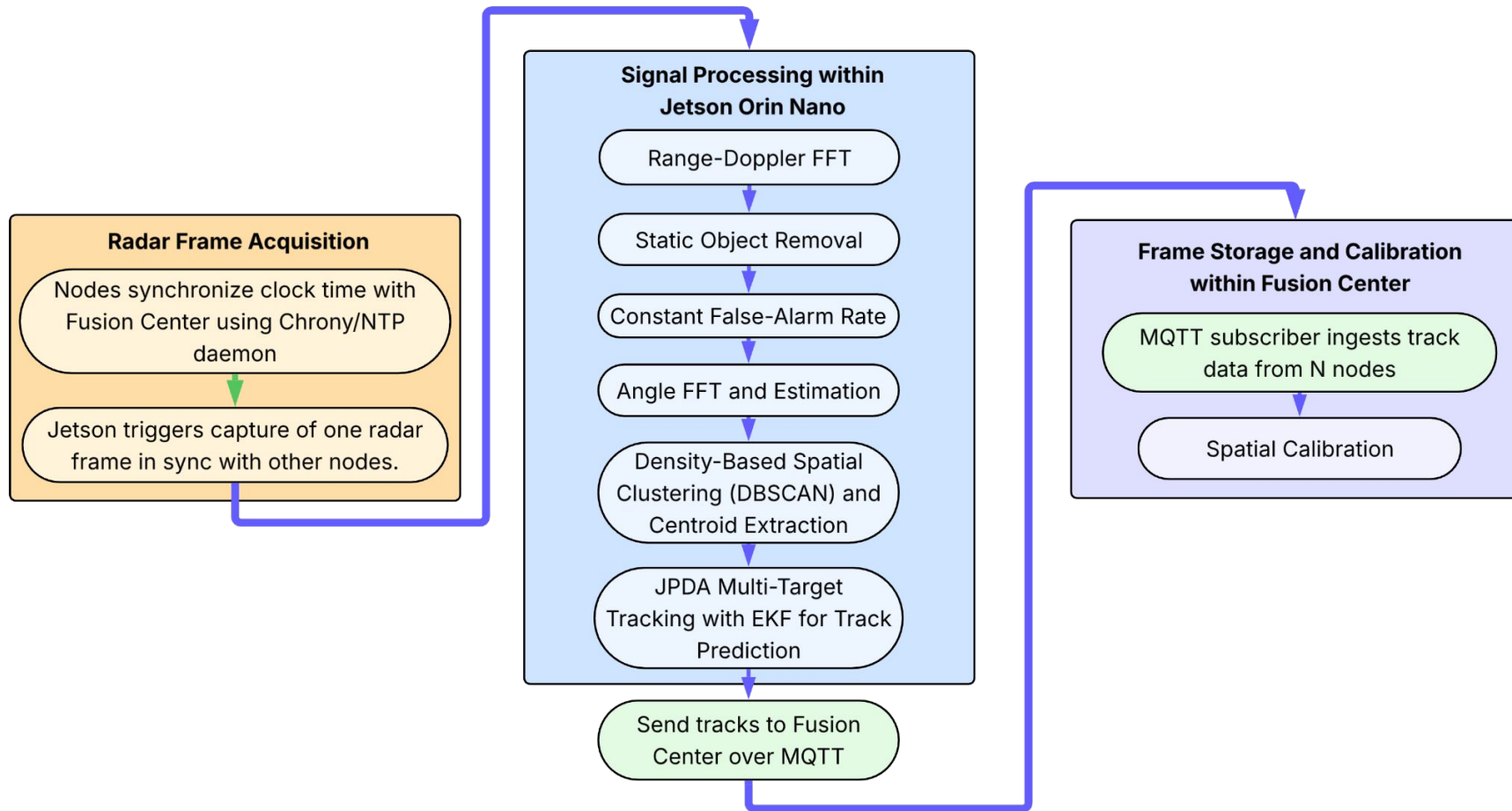


Pipeline: Filtering Centroid Noise

- Filter Centroids across time using Extended Kalman Filtering (EKF) and Joint Probabilistic Data Association (JPDA)

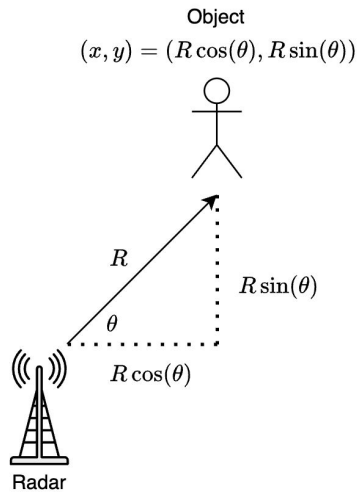
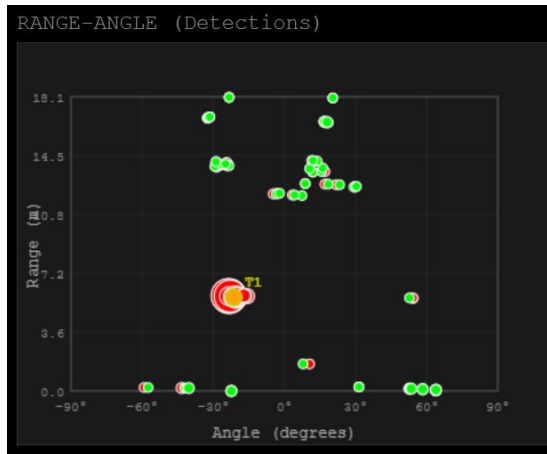


Dataflow Overview: Spatial Calibration



Spatial Calibration

- Goals:
 - Find out what rotation + translation turns node B's coordinates into node A's coordinates?
 - Get a 2D bird's eye map to visualize the fused trajectories
- Process:
 1. Nodes publish frames whose payloads carry detections including range and angle
 2. Source only 'common frames' where we have data from all nodes
 3. Turn each frame's detection list into centroids
 - Mean range across detections in that frame
 - Circular mean angle (b/c plain averaging breaks near $\pm\pi$ wrap)
 - Convert to Cartesian
 4. Estimate rotation between node angles using
 - Mean-removal
 - complex correlation phase
 5. Estimate translation between nodes via mean adjustment
 6. Use translation and rotation data to transpose each node's centroids to create bird's eye map



Video Demo: needs to be edited



Thank you for listening!
Feel free to ask any questions.

Special thanks to: Anirban Banik, Tien Nguyen, Prof. Isukapalli, Prof. Madhow